

SCIENCE WRITING EXPERIENCE

American Astronomical Society

Communications Manager, AAS Nova Editor, Deputy Press Officer	2026 – Present
Communications Specialist, AAS Nova Editor, Deputy Press Officer	2021 – 2026

ESA/Hubble and ESA/Webb Outreach Team

Science Writer	2024 – Present
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Voice of America

AAAS Mass Media Science and Engineering Fellow	2019
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AAS Nova

AAS Media Fellow	2018 – 2019
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Astrobits

Author and Editor	2017 – 2018
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EDUCATION

Boston University, Boston, MA

PhD, Astronomy, September 2021

MA, Astronomy, May 2017

Williams College, Williamstown, MA

BA, Astrophysics (with honors) and Chinese, *cum laude*, June 2014

RESEARCH EXPERIENCE

Graduate Research Assistant

<i>Boston University; Dr. Paul Withers</i>	2016 – 2021
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Sally Ride Undergraduate Research Intern

<i>Jet Propulsion Laboratory; Dr. Bonnie Buratti</i>	2014
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Undergraduate Research Assistant

<i>Williams College; Dr. Karen Kwitter</i>	2013 – 2014
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HONORS & AWARDS

Venus Exploration and Analysis Group (VEXAG) Travel Grant	2019
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Future Investigators in NASA Earth and Space Science and Technology (FINESST) Grant	2019
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Massachusetts Space Grant Consortium Graduate Fellowship	2019
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AAS Media Fellowship	2018
Ewha-Luce International Seminar	2017
Clare Boothe Luce Graduate Fellowship	2016
Departmental Honors in Astrophysics	2014
Fulbright English Teaching Assistantship	2014

REFEREED PUBLICATIONS

Withers, P., Felici, M., **Hensley, K.**, Mendillo, M., Barbinis, E., Kahan, D., Oudrhiri, K., & Girazian, Z. (2023) The Ionosphere of Mars from Solar Minimum to Solar Maximum: Dayside Electron Densities from Mavén and Mars Global Surveyor Radio Occultations, *Icarus*. doi:10.1029/2022JA030998

Hensley, K., Withers, P., & Thiemann, E. (2022) Composition of Mars's Dayside Ionosphere Under Changing Solar Irradiance Conditions, *Journal of Geophysical Research: Space Physics*. doi:10.1029/2022JA030998

Felici, M., Withers, P., Vogt, M.F., **Hensley, K.**, & Andersson, L. (2022) Electron Densities in the Ionosphere of Mars: Comparison of MAVEN/ROSE and MAVEN/LPW Measurements, *Journal of Geophysical Research: Space Physics*. doi:10.1029/2021JA030155

Withers, P., Felici, M., **Hensley, K.**, Mendillo, M., Oudrhiri, K., Kahan, D., & Girazian, Z. (2021) The Ionosphere of Mars from Solar Minimum to Solar Maximum: Dayside Electron Densities from MAVEN and Mars Global Surveyor Radio Occultations, *Icarus*. doi:10.1016/j.icarus.2021.114508

Hensley, K. & Withers, P. (2021) Response of Mars's Topside Ionosphere to Changing Solar Activity and Comparisons to Venus, *Journal of Geophysical Research: Space Physics*. doi:10.1029/2020JA028913

Withers, P., **Hensley, K.**, Vogt, M., & Hermann, J. (2020) Recovery and Validation of Venus Ionospheric Electron Density Profiles from Pioneer Venus Orbiter Radio Occultation Observations, *Planetary Science Journal*, 1, 78. doi:10.3847/PSJ/abcaf9 [pdf]

Withers, P., **Hensley, K.**, Vogt, M., & Hermann, J. (2020) Recovery and Validation of Venus Neutral Atmospheric Density Profiles from Pioneer Venus Orbiter Radio Occultation Observations, *Planetary Science Journal*, 1, 79. doi:10.3847/PSJ/abc476 [pdf]

Hensley, K., Withers, P., Girazian, Z., Paetzold, M., Tellmann, S., & Hausler, B. (2020) Dependence of Dayside Electron Densities at Venus on Solar Irradiance, *Journal of Geophysical Research: Space Physics*, 125, 2. doi:10.1029/2019JA027167 [pdf]

Corradi, R.L.M., Kwitter, K.B., Balick, B., Henry, R.B.C., & **Hensley, K.** (2015) The Chemistry of Planetary Nebulae in the Outer Regions of M31, *Astrophysical Journal*, 807, 181. doi:10.1088/0004-637X/807/2/181 [pdf]

INVITED TALKS & SELECTED PRESENTATIONS

SOLAR IRRADIANCE EFFECTS ON IONOSPHERIC COMPOSITION AT VENUS AND MARS
University of Kansas Astronomy and Space Physics Seminar Series, Oct. 2021

FROM SCIENCE TO SCICOMM
Li, Nishimura, and Walsh Research Group Meeting, Apr. 2021

DAYSIDE ION COMPOSITION AT MARS UNDER CHANGING SOLAR IRRADIANCE
MAVEN Project Science Group Meeting, Virtual, Apr. 2021

WILLIAMS COLLEGE PHYSICS AND ASTRONOMY COLLOQUIUM SERIES
Canceled due to COVID-19

A TALE OF TWO PLANETS: IONOSPHERES OF VENUS AND MARS
Boston University Student Seminar, Boston, MA, Nov. 2020

SOLAR ACTIVITY VARIATIONS OF THE IONOSPHERES OF VENUS AND MARS
American Geophysical Union Fall Meeting, San Francisco, CA, Dec. 2019

VARIATIONS IN THE TOPSIDE VENUS IONOSPHERE
Mars/Venus Express Radio Science Team Meeting, Truckee, CA, Oct. 2017

LIVING WITH A STAR: FROM THE SUN TO PROXIMA CENTAURI
BU Center for English Language and Orientation Programs, Boston, MA, Feb. 2019
BU Center for English Language and Orientation Programs, Boston, MA, Aug. 2018

TERRESTRIAL PLANET IONOSPHERES & EMERGING ISSUES IN ASTRONOMY
Ewha-Luce International Seminar, Seoul, South Korea, July 2017

OUTREACH, MENTORING, & SERVICE

Astronomy Department Representative 2019 – 2020
Served as the department representative to BU's Graduate Student Organization.

Letters to a Pre-Scientist Pen Pal 2018 – 2020
Exchanged letters with middle school students. The goal of Letters to a Pre-Scientist is to demystify science and make it more accessible, especially to students from low-income backgrounds who may have less exposure to scientists.

BU Graduate Women in Science and Engineering Mentor 2016 – 2017
Served as an academic and professional development mentor for Boston University undergraduate biomedical engineering student Xiaoshan Ke.

BU Research in Science & Engineering (RISE) Mentor 2016
Served as a research mentor for high school student Arthur Chen on a project that used radio occultations to map the atmosphere of Venus.

SCIENCE WRITING

AAS Nova [>100 articles]	2018 – 2019, 2021 – Present
<i>Sky & Telescope</i> Magazine	2021
<i>Highland Outdoors</i> Magazine [Wild, Wonderful, and Radio Quiet]	2021
The College Board	2020 – 2021
Voice of America [12 articles]	2019
Astrobites [18 articles]	2017 – 2018
Baen Books Free Nonfiction [Small Stars ; Plasma Frequency]	2018, 2019

PODCAST APPEARANCES

Best Virginia S2E10 Is There Anybody Out There?	2022
astro[sound]bites Episode 43. Welcome to the World of Science Communication	2021